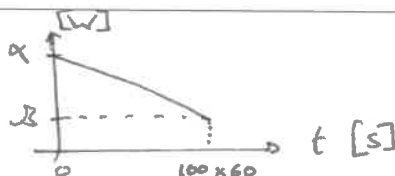


Section 1 / Afdeling 1

Question 1 [2]

- 01 A 9 V battery is used to power a remote-controlled car. The power delivered by the battery moves linearly from α W at $t = 0$ min to β W at $t = 100$ min. Determine the total energy used by the car.
(The values of α and β are given in ClickUP.)

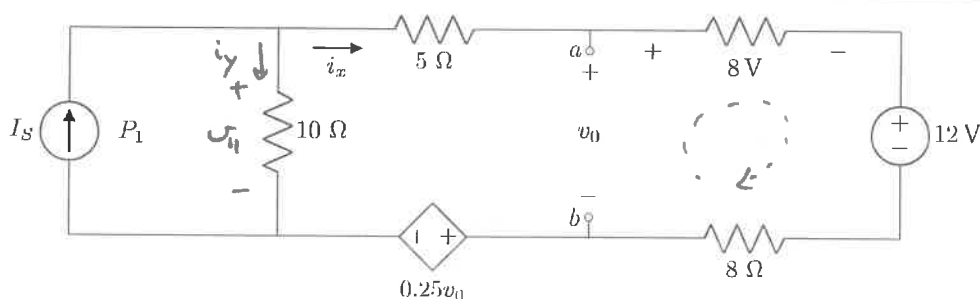
$$w = \int_0^t p \, dt$$



$$w = \text{Area under curve} \\ = \left(\beta + \frac{1}{2}(\alpha - \beta) \right) \times 100 \times 60.$$

Questions 2 to 3 [4]

(The values of I_S and i_x are given in ClickUP.)



- 02 Calculate the power associated with the component labelled as P_1 . [2]
03 Determine v_0 . [2]

02 • $i_y = I_S - i_x$
• $v_{11} = 10 i_y = 10(I_S - i_x)$
• $\therefore P_1 = -v_{11} \times I_S = -10(I_S - i_x) I_S.$

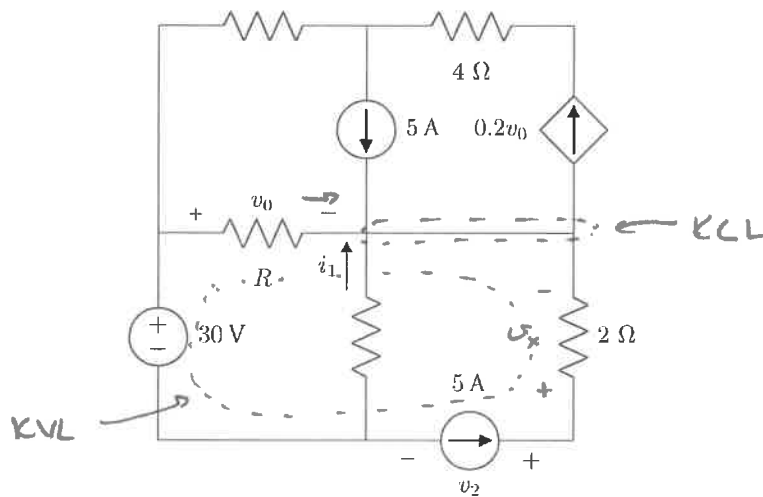
03 • KVL: $-v_0 + 8 + 12 + 8 i_x = 0$
 $v_0 = 20 + 8 i_x$

check:

KVL: $5 i_x + v_0 + 0.25 v_0 - v_{11} = 0$
 $1.25 v_0 = 10(I_S - i_x) - 5 i_x$

$$v_0 = 8 I_S - 12 i_x$$

Questions 4 to 5 [4]

Determine i_1 and v_2 .(The values of v_0 and R are given in ClickUP.)04 i_1 [2]05 v_2 [2]05 v_2 *

$$v_x = 2 \times 5 = 10V$$

$$* \text{ KVL: } -30 + v_0 - v_x + v_2 = 0$$

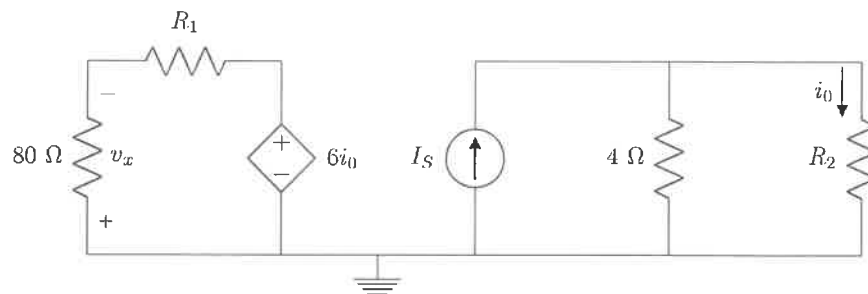
$$\therefore v_2 = 30 + 10 - v_0 = \underline{40 - v_0}$$

06 i_1 *

$$* \text{ KCL: } i_1 + 5 + \frac{v_0}{R} + 5 = 0.2v_0.$$

$$i_1 = \underline{\left(0.2 - \frac{1}{R}\right) v_0 - 10.}$$

Questions 6 to 7 [4]

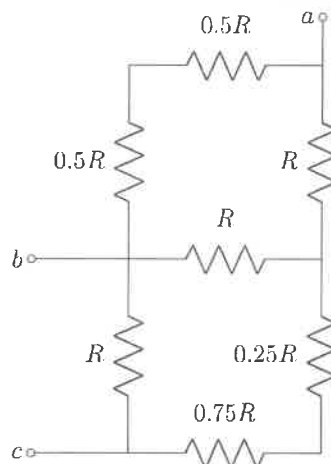
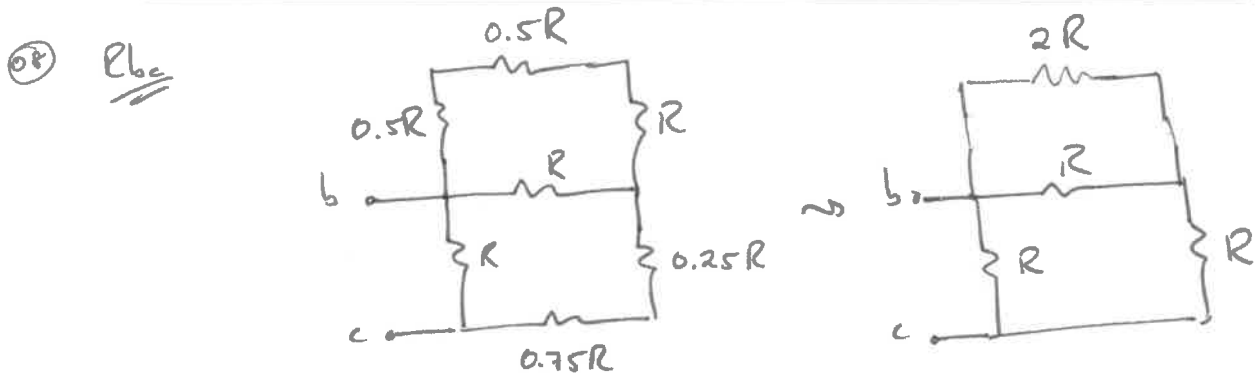
Determine I_S and v_x .(The values of i_0 , R_1 and R_2 are given in ClickUP.)06 v_x [2]07 I_S [2]

$$\textcircled{07} * \quad i_0 = \frac{4 \times I_S}{4 + R_2} \quad (\text{Current division})$$

$$\therefore I_S = \frac{(4 + R_2) \times i_0}{4} = (1 + 0.25 R_2) i_0$$

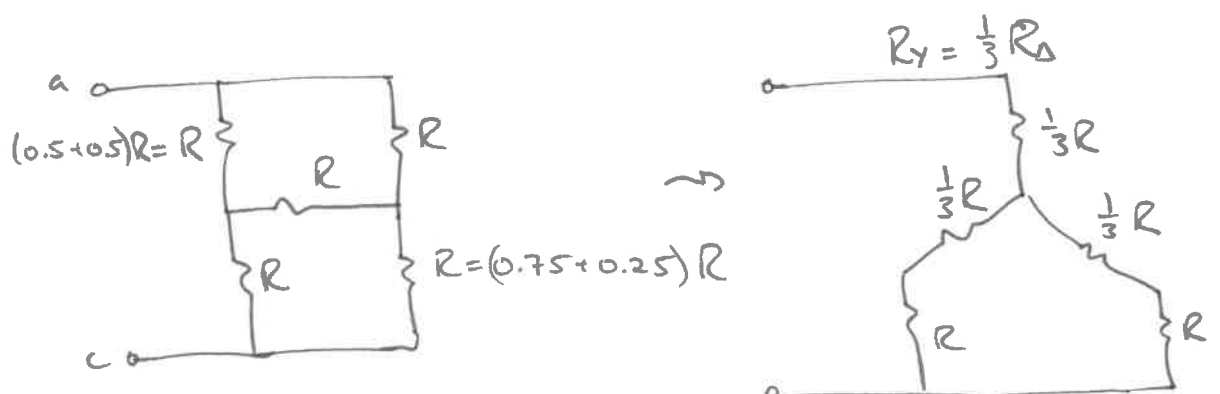
$$\textcircled{06} * \quad v_x = \frac{-(6i_0) \times 80}{80 + R_1} = \frac{-480i_0}{80 + R_1} \quad (\text{Voltage division})$$

Questions 8 to 9 [4]

(The value of R is given in ClickUP.)08 Determine R_{bc} . [2]09 Determine R_{ac} . [2]

$$\therefore R_{bc} = R \parallel (2R \parallel R + R)$$

$$= R \parallel \left(\frac{2R}{3} + R \right) = \underline{\underline{\frac{5}{8} R}}$$

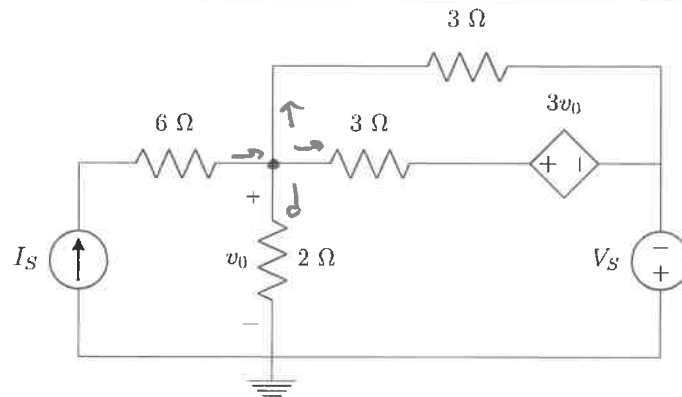
09 R_{ab} 

$$\therefore R_{ab} = \frac{1}{3}R + \left(\frac{1}{3}R + R \right) \parallel \left(\frac{1}{3}R + R \right)$$

$$= \frac{1}{3}R + \frac{2}{3}R$$

$$= \underline{\underline{R}}$$

Question 10 [2]

(The values of V_S and I_S are given in ClickUP.)10 Determine the voltage v_0 .Nodal analysis

$$I_S = \frac{v_0}{2} + \frac{v_0 - (-V_S)}{3} + \frac{v_0 - (3v_0) - (-V_S)}{3}$$

$$I_S = \frac{v_0}{2} + \frac{v_0 + V_S}{3} + \frac{v_0 - 3v_0 + V_S}{3}$$

$$(\times 6) \quad 6I_S = 3v_0 + 2v_0 + 2V_S + 2v_0 - 6v_0 + 2V_S$$

$$v_0 = \underline{6I_S - 4V_S}$$

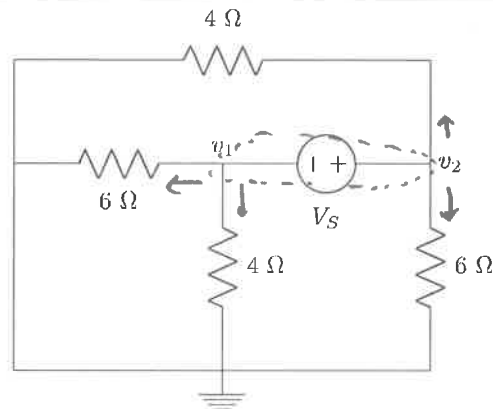
Question 11 [2]

Nodal analysis can be used to solve for the two node voltages shown below. The circuit is an example of a supernode. The KVL equation for the supernode is as follows:

$$-v_1 + v_2 = V_S$$

Setup the KCL equation for the supernode and solve for v_2 .

(The value of V_S is given in ClickUP.)



11 v_2 [2]

$$\text{KCL: } \frac{v_1}{4} + \frac{v_1}{6} + \frac{v_2}{6} + \frac{v_2}{4} = 0.$$

$$(\times 12) \quad 3v_1 + 2v_1 + 2v_2 + 3v_2 = 0$$

$$5v_1 = -5v_2$$

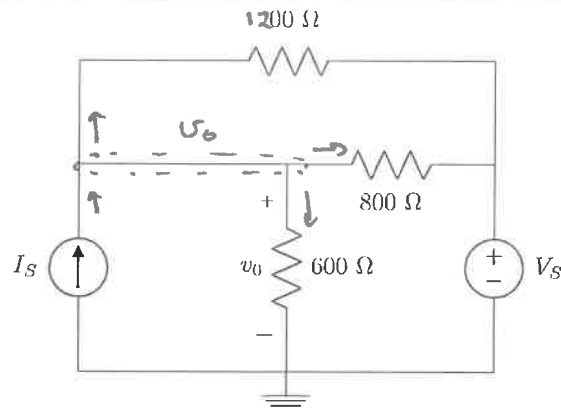
$$v_1 = -v_2$$

$$\text{Use } -v_1 + v_2 = V_S.$$

$$\therefore -(-v_2) + v_2 = V_S$$

$$v_2 = V_S/2$$

Question 12 [2]

(The values of V_S and I_S are given in ClickUP).12 Determine v_0 [2]* Nodal analysis

$$I_S = \frac{v_0 - V_S}{1200} + \frac{v_0 - V_S}{800} + \frac{v_0}{600}$$

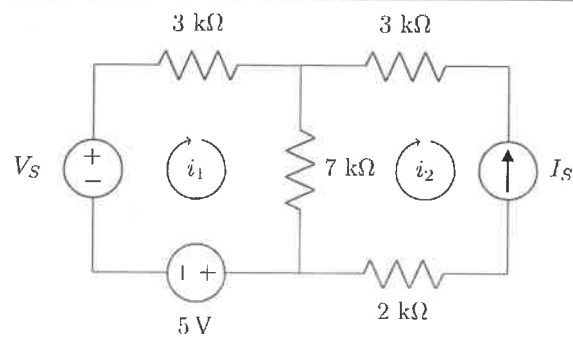
$$(\times 2400) \quad 2400 I_S = 2v_0 - 2V_S + 3v_0 - 3V_S + 4v_0$$

$$v_0 = 2400 I_S + 5V_S$$

$$v_0 = \frac{2400 I_S + 5V_S}{1}$$

→

Question 13 [2]

(The values of V_S and I_S are given in ClickUP).13 Determine i_1 [2]

$$i_2 = -I_S$$

$$+5 - V_S + 3k i_1 + 7k i_1 - 7k i_2 = 0.$$

$$10k i_1 = V_S - 5 + 7k i_2$$

$$i_1 = \frac{V_S - 5 + 7k i_2}{10k.}$$

—————→

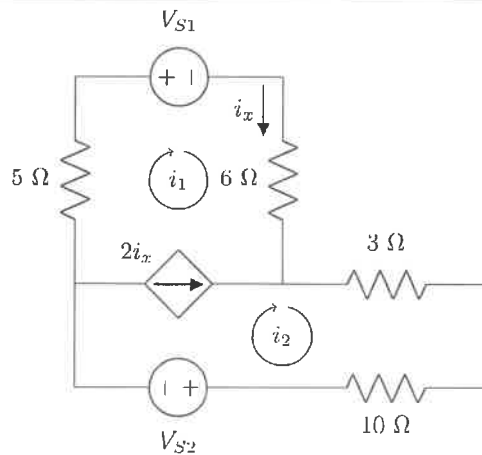
Question 14 [2]

The circuit below is an example of a supermesh. The following is known:

$$-i_1 + i_2 = 2i_x \quad (\text{KCL equation for dependent current source } 2i_x)$$

Using this information, setup the KVL supermesh equation and solve for i_x .

(The values of V_{S1} and V_{S2} are given in ClickUP).



14

 i_x [2]

$$* \quad -i_1 + i_2 = 2i_x$$

$$\text{but } i_1 = i_x$$

$$\therefore -i_x + i_2 = 2i_x$$

$$i_2 = 3i_x$$

$$\text{Supermesh } * \quad 13i_2 + V_{S2} + 11i_1 + V_{S1} = 0$$

$$\therefore 39i_x + 11i_x = -V_{S1} - V_{S2}$$

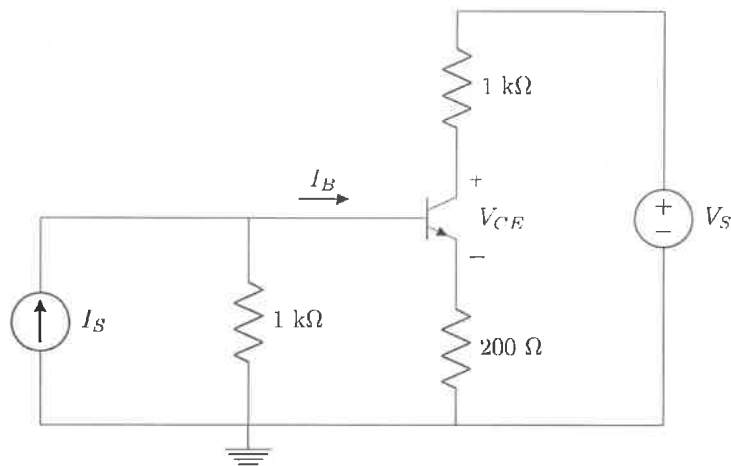
$$i_x = - \frac{(V_{S1} + V_{S2})}{50}$$

50. Ω

Questions 15 to 16 [4]

It is known that $\beta = 99$ and $V_{BE} = 0.7$ V. Determine I_S and V_{CE} .

(The values of I_B and V_S are given in ClickUP.)



15 I_S [2]

16 V_{CE} [2]

★ Source transform:



(15) KVL $\rightarrow - (1kI_S) + 1kI_B + 0.7 + (\beta + 1)I_B(200) = 0.$

$$I_S = \frac{1kI_B + 0.7 + 100 \times 200 I_B}{1k} = 21I_B + \frac{0.7}{1000}$$

(16) KVL $\rightarrow -V_S + (\beta I_B)1k + V_{CE} + (\beta + 1)I_B(200) = 0$

$$V_{CE} = V_S - 99kI_B - 100 \times 200 I_B$$

Total Section 1 / Totaal Afdeling 1: [32]